Prayas JEE AIR (2025)

Advanced

Weekly Test 22

DURATION: 180 Minutes | **DATE : 21/12/2024** | **M. MARKS: 198**

Topics covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 54 questions. The maximum marks are **198**.

SECTION 1 (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases.

SECTION 2 (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is(are) chosen;

Partial Marks : + 3 If all the four options are correct but ONLY three options are chosen;

Partial Marks : + 2 If three or more options are correct but ONLY two options are chosen, both of which are correct;

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : –2 In all other cases.

SECTION 3 (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered;

Zero Marks : 0 In all other cases.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Students (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ Invigilator's Signature _____

IMPORTANT CONSTANTS

Speed of light in free space,	:	$3.00 \times 10^{8}\ ms^{-1}$
Permeability of free space,	:	$4\pi \times 10^{-7}\ Hm^{-1}$
Permittivity of free space,	:	$8.85 \times 10^{-12}\ Fm^{-1}$
The Planck constant,	:	$6.63 \times 10^{-34}\ Js$
Rest mass of electron,	:	$9.1 \times 10^{-31}\ kg$
Rest mass of proton,	:	$1.67 \times 10^{-27}\ kg$
Molar gas constant,	:	$8.31\ JK^{-1}\ mol^{-1}$
The Avogadro constant,	:	$6.02 \times 10^{23}\ mol^{-1}$
The Boltzmann constant,	:	$1.38 \times 10^{-23}\ JK^{-1}$
Gravitational constant,	:	$6.67 \times 10^{-11}\ N\ m^{2}kg^{-2}$
Acceleration of free fall	:	$9.8\ ms^{-2}$
Rydberg Constant	:	$1.097 \times 10^{7}\ m^{-1}$
Atomic mass unit	:	$1.67 \times 10^{-27}\ kg$
Charge on proton	:	$1.6 \times 10^{-19}\ C$

IMPORTANT VALUES

$\sqrt{2}$ = 1.414

$\sqrt{3}$ = 1.732

$\sqrt{5}$ = 2.236

π = 3.142

e (Euler's constant) = 2.718

ln 10 = 2.303

$\log_{10}2$ = 0.3010

$\log_{10}3$= 0.4770

$\log_{10}7$ = 0.845

**** Use above values unless otherwise specified in a question.***

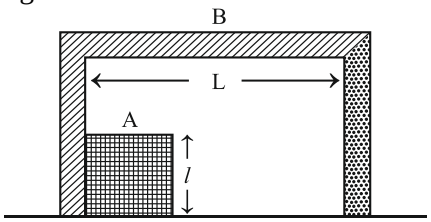
❑❑❑

PART-I (PHYSICS)

SECTION-1

Single Correct Type Questions:

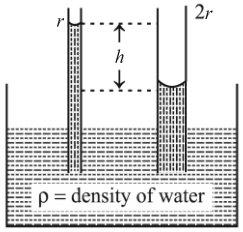
1. A cubical block *A* of mass *m* and a box *B* of same mass are lying on a horizontal surface as shown in the above figure. The coefficient of kinetic friction between the block and horizontal surface is μ_A, while that between the box and surface is μ_B. Given that $\mu_B > \mu_A$. At $t = 0$, the position is as shown in the figure and both are simultaneously given a velocity v_0 to the right. If the collision between the block and the box takes place, then the time after which block will collide with the right end of the box is



(A) $\sqrt{\dfrac{2(L-l)}{(\mu_B-\mu_A)g}}$ (B) $\sqrt{\dfrac{2(L-l)}{(\mu_B+\mu_A)g}}$

(C) $\sqrt{\dfrac{(L+l)}{2(\mu_B-\mu_A)g}}$ (D) $\sqrt{\dfrac{(L-l)}{(\mu_B-\mu_A)g}}$

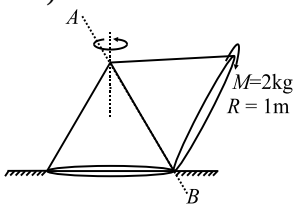
2. In the given arrangement, two capillaries of glass having radii *r* and 2*r* are dipped in pure water of density ρ, the surface tension of water is (contact angle is equal to zero)



(A) $2\rho hgr$ (B) ρhgr

(C) $\dfrac{\rho hgr}{2}$ (D) $4\rho hgr$

3. A hollow cone (*M*, *R*) of semi apex-angle 30° is rolling without slipping on a fixed cone (same radius and semi-angle) as shown. If angular speed of rotation is 2 rad/s, find energy due to motion of the cone. (Given that moment of inertia of a hollow cone about its slant axis *AB* is $I = \dfrac{MR^2}{4}\left(1+3\cos^2(\alpha)\right)$, where α is semi apex-angle of cone)



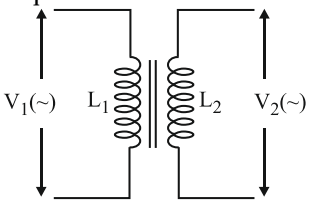
(A) $\dfrac{13}{4}$ J

(B) 6 J

(C) $\dfrac{8}{3}$ J

(D) $\dfrac{39}{4}$ J

4. For an ideal transformer shown in the figure, ratio of V_1 & V_2 is equal to



(Where L_1 & L_2 are self-inductances of primary and secondary windings. For an ideal transformer coefficient of mutual induction for pair of coils is $M = \sqrt{L_1 L_2}$):

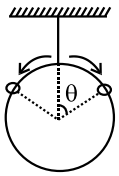
(A) $\dfrac{L_1}{L_2}$

(B) $\sqrt{\dfrac{L_1}{L_2}}$

(C) $\dfrac{L_2}{L_1}$

(D) $\sqrt{\dfrac{L_2}{L_1}}$

5. A fixed smooth circular track of mass *M* is vertically hung by a string down the ceiling. Two small rings, each of mass *m* are initially at the top of the track. They slide down simultaneously along the track in opposite directions. Find the position of the rings, when tension in the string becomes zero for the first time.



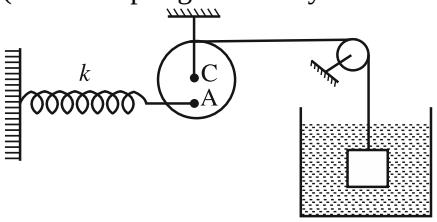
(A) $\theta = \cos^{-1}\left[\dfrac{5}{3}\left(1+\sqrt{1-\dfrac{3M}{2m}}\right)\right]$

(B) $\theta = \sin^{-1}\left[\dfrac{5}{3}\left(1+\sqrt{1-\dfrac{3M}{2m}}\right)\right]$

(C) $\theta = \cos^{-1}\left[\dfrac{1}{3}\left(1+\sqrt{1-\dfrac{3M}{2m}}\right)\right]$

(D) $\theta = \sin^{-1}\left[\dfrac{1}{3}\left(1+\sqrt{1-\dfrac{3M}{2m}}\right)\right]$

6. A cubical block of volume V and density 3ρ is placed inside a liquid of density ρ and attached to a spring of spring constant k via an inextensible massless string and pulley as shown in the figure. Assuming spring and pulley to be ideal and spring is attached at A which is at $R/2$ from centre. The compression in the spring at equilibrium is (Assume Spring will always remain horizontal)



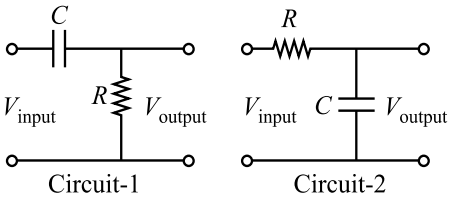
(A) $\frac{4\rho Vg}{k}$ (B) $\frac{2\rho Vg}{k}$

(C) $\frac{3\rho Vg}{k}$ (D) $\frac{\rho Vg}{k}$

SECTION-2

One or More Than One Correct Type Questions:

7. Figure shows two circuits (i.e., circuit 1 and circuit 2). In each case V_0 denotes peak value of input voltage. Input voltage is given by $V_{input} = V_0 \sin \omega t$ and output voltage by $V_{output} = V_M \sin(\omega t + \delta)$



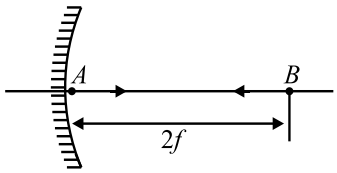
(A) In circuit-1, maximum output voltage is $V_M = \frac{V_0}{\sqrt{1+(\omega RC)^{-2}}}$

(B) In circuit-2, maximum output voltage is $V_M = \frac{V_0}{\sqrt{1+(\omega RC)^{2}}}$

(C) In circuit-1, average power dissipated is given by $P_{avg} = \frac{V_0^2}{2R\left[1+(\omega RC)^{-2}\right]}$

(D) Phase angle δ for circuit-2 is given by $|\tan\delta| = \omega RC$

8. At time $t = 0$, two point objects A and B respectively are at pole and centre of curvature of a fixed concave mirror of focal length f, the velocity vectors of A and B are always $\vec{v}_A = u\hat{i}$ and $\vec{v}_B = -u\hat{i}$ respectively, where $\hat{i}$ is unit vector along principal axis directed from pole towards focus and u is a positive constant



(A) The distance between images of A and B will be $4f$ at time $t = \frac{f}{2u}$

(B) Magnitude of relative velocity of image of A and image of B at $t = 0$ is $2u$

(C) Starting from $t = 0$ and before the particles come in contact, distance between image A and B increases

(D) Starting from $t = 0$ and before the particles come in contact, distance between A and B first increases and then decreases

9. A spherical balloon contains 1 mole of He at T_0. The balloon material is such that the pressure inside is always proportional to square of diameter. When volume of balloon becomes 8 times of initial volume, (R is universal gas constant) then:-

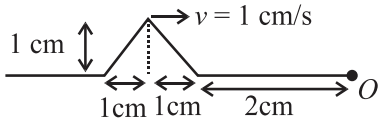
(A) Work done by gas $\frac{93RT_0}{5}$

(B) Final temperature of the gas is $32\ T_0$

(C) Heat taken by gas $65.1\ RT_0$.

(D) Heat given by gas $65.1\ RT_0$.

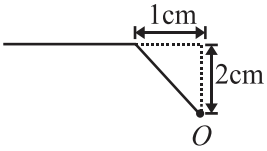
10. A wave pulse on a string has the dimension shown in figure. The wave speed is $v = 1$ cm/s. If point O is a free end, choose the correct option(s)



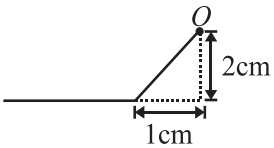
(A) The wave pulse suffers a phase change of π on reflection

(B) The wave pulse does not suffer any phase change on reflection

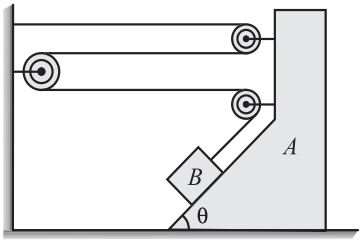
(C) The shape of the wave at $t = 3$ s is



(D) The shape of the wave at $t = 3$ s is



11. In the diagram shown, the acceleration of the block B as shown in figure relative to the block A and relative to ground is a_{BA} and a_{BG}, respectively. If the block A is moving toward left with an acceleration a_0, then



(A) $a_{BA} = 2a_0$ (B) $a_{BG} = 3a_0$

(C) $a_{BA} = 3a_0$ (D) $a_{BG} = a_0\sqrt{10 + 6\cos\theta}$

12. A particle is projected with a velocity $2\sqrt{hg}$ so that it just clears two walls of equal height h at horizontal separation $2h$ from each other. Then the

(A) angle of projection is 30° with vertical.

(B) angle of projection is 30° with horizon.

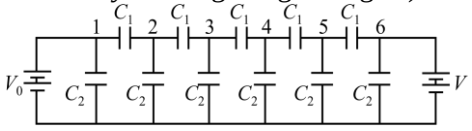
(C) time of passing between the walls is $\sqrt{\frac{2h}{g}}$.

(D) time of passing between the walls is $2\sqrt{\frac{h}{g}}$.

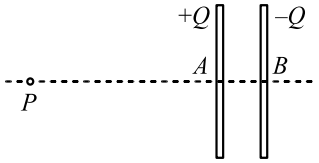
SECTION-3

Numerical Value Type Questions:

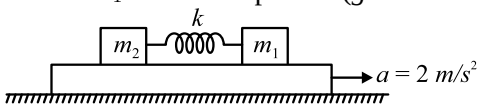
13. In the circuit shown, ratio of capacitors is $C_2/C_1 = 4/3$ and terminal 'voltage of the battery connected at the left end is V_0. If the terminal voltage of the battery V connected at the right most end, in such a way that the potentials of nodes 1, 2, 3, 4, 5 and 6 are in a geometric progression, is of form $V = V_0x^y$, then the value of $|x| + |y|$ is:
(**Note:** x & y are single digit integers)



14. Two identical thin dielectric plates each of area S carrying uniform charges $+Q$ and $-Q$ are placed parallel to each other. Both the plates have small holes A and B at their centers. Distance d between the plates and size of holes are so small as compared to their linear dimensions that electric field between them can be assumed almost uniform and fringing of field near the edges can be ignored. A particle P of mass m and charge $-q$ is placed on the line joining the holes A and B at a great distance as shown in the figure. It is projected towards the capacitor with a velocity that is twice of the minimum velocity required to pass through the capacitor. The particle exits the capacitor with a speed equal to $\sqrt{\frac{nqQd}{\varepsilon_0 mS}}$. Calculate the value of n. (Q and q are positive quantities)

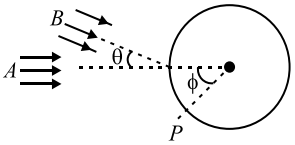


15. Two blocks of masses m_1 = 1 kg and m_2 = 2 kg are connected by a spring of constant k = 100 N/m and placed over a plank as shown. The coefficient of friction between m_2 and plank is 0.5 and there is no friction between m_1 and plank. Initially the whole system, is at rest. The plank is now pulled to right with constant acceleration a = 2 m/s^2. Find the amplitude (in cm) of oscillation of block m_1 relative to plank. (g = 10 m/s^2)



16. A stretchable conducting ring is in the shape of a circle. It is kept in a uniform magnetic field (B) that is perpendicular to the plane of the ring. The ring is pulled out uniformly from all sides so as to increase its radius at a constant rate $\frac{dr}{dt} = V$ while maintaining its circular shape. Resistance of the ring remains constant at R. If the rate of work done by the external agent against the magnetic force when the radius of the ring is r_0 is $\frac{x\pi^2 B^2 r_0^2 V^2}{R}$. Find x.

17. Two identical beams A and B of plane coherent waves of the same intensity and wavelength λ fall on cylindrical screen. The angle between the directions of the beam propagations is θ. Consider a point P on the screen at angular position ϕ from the beam A as shown in the figure. Find the closest integer to distance in nanometers between adjacent interference fringes on the screen near the point P. Assume the distance β between adjacent fringes is much less than the radius of the cylinder. (Given: θ = 60°, ϕ = 15°, λ = $\sqrt{2}$ nm)



18. An ice cube of mass 0.1 kg at 0°C is placed in an isolated container which is at 227°C. The specific heat c of the container varies with temperature T according to the empirical relation $c = A + BT$, where A = 100 cal kg^{-1} K^{-1} and $B = 2 \times 10^{-2}$ cal kg^{-1} K^{-2}. If the final temperature of the container is 27°C, the mass of the container is $\frac{107}{n}$ kg, the value of n is (Latent heat of fusion for water = 8×10^4 cal kg^{-1}, specific heat of water = 10^3 cal kg^{-1} K^{-1})

PART-II (CHEMISTRY)

SECTION-1

Single Correct Type Questions:

19. Which one of the following statements is not true for $IO_2F_2^-$ according to VSEPR theory?

(A) The lone pair and two I-O double bonds occupy the equatorial positions of trigonal bipyramid

(B) It has sp^3d hybridization and is See-Saw shaped

(C) Its structure is analogous to SF_4

(D) Only one of the F atoms occupy equatorial position

20. In which one of the following pairs, first complex can show geometrical isomerism and second complex can show optical isomerism?

(A) $\left[Cr(NH_3)_5Cl\right]Cl_2, \left[Cr(gly)_3\right]$

(B) $\left[Cr(H_2O)_2(NH_3)_4\right]Cl_3, \left[Cr(NH_3)_3Cl_3\right]$

(C) $\left[Cr(NH_3)_3Cl_3\right], \left[Cr(en)_3\right]Cl_3$

(D) $\left[Cr(gly)_3\right], \left[Cr(NH_3)_5Cl\right]Cl_2$

21. 1 mole of $H_2(g)$ and 1 mole of $I_2(g)$ are taken in a container at 400 K and the following equilibrium is allowed to establish:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

The equilibrium mixture required 100 mL of 10 M hypo solution upto starch end point (i.e., blue colour of starch indicator is disappeared). The dissociation constant per mole of HI at 400 K for reaction $HI_{(g)} \rightleftharpoons \frac{1}{2}H_{2(g)} + \frac{1}{2}I_{2(g)}$ is

(A) 0.25 (B) 0.50

(C) 4 (D) 2

22. The incorrect statement among the following is

(A) Ce^{4+}, Yb^{2+} and Lu^{3+} all are diamagnetic ions.

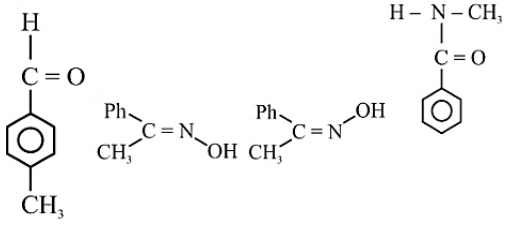
(B) Carbides of lanthanoids produce C_2H_2 on hydrolysis

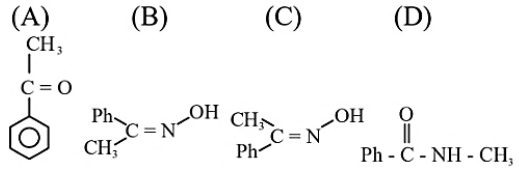
(C) Lanthanoids are coloured due to f-f transition

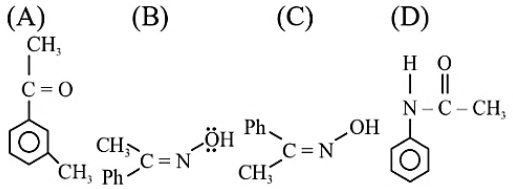
(D) All lanthanoids are radioactive except Pm

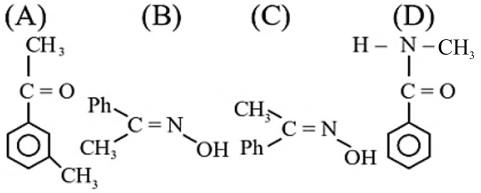
23. An aromatic organic compound (A) C_8H_8O gives positive iodoform test but negative Tollen's test. A on reaction with NH_2OH forms B and C that are geometrical isomers. B and C on reaction with H_2SO_4 forms a compound D and E respectively. In compound D , electrophile attack occurs on meta position of ring while in E electrophile attack occurs on ortho and para positions. The correct representations of structures are:

(A) (B) (C) (D)

(A) 

(B) 

(C) 

(D) 

24. Which one the following oxides gives two oxyacids in water?

(A) N_2O_3

(B) N_2O_5

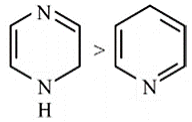
(C) NO_2

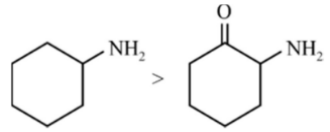
(D) N_2O

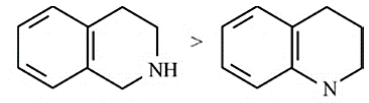
SECTION-2

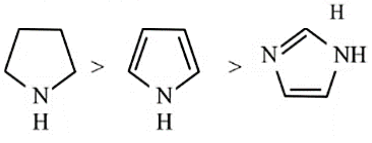
One or More Than One Correct Type Questions:

25. Which of the following orders for basic strength is/are correct?

(A) 

(B) 

(C) 

(D) 

26. Select the correct statement(s) among the following

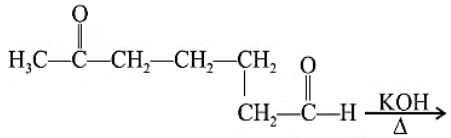
(A) Chelation effect is maximum for five and six membered rings

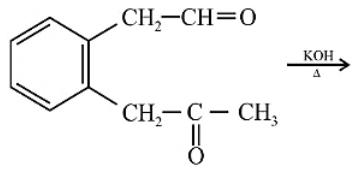
(B) Greater the charge on the central metal cation, greater will be the value of Δ (CFSE) in magnitude.

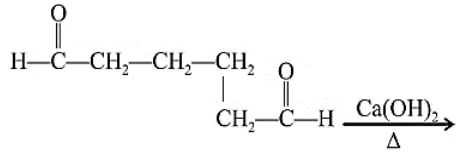
(C) In complex ion $[CoF_6]^{3-}$, F^- is a weak field ligand, so that $\Delta_0 < P$ (Pairing energy) and it is low spin complex.

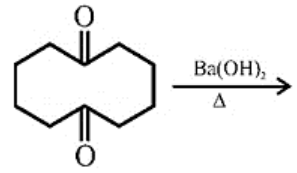
(D) $[CoCl_2(NH_3)_2\ (en)]^+$ complex ion will have four different isomers.

27. In which of the following intramolecular aldol condensation reaction(s), more than two products (stereoisomers are not considered) are possible?

(A) 

(B) 

(C) 

(D) 

28. In the crystal of solid (x), A^{3+} ions are present at each of the corners and B^- ions are present at each of the face centres of a cubic unit cell. When 0.6 g of the above solid is dissolved in 100 g of water, the solution showed a freezing point of –0.2976° C. The solute (x) is assumed to be 20% dissociated in aqueous solution and K_f of H_2O is 1.86 K kg mol^{-1}.

(Given that $r_{A^{3+}} = 0.414$ Å, $r_{B^-} = 1.0$ Å and assume that element 'A' exists in +3 and +4 oxidation states and Avogadro's number = 6×10^{23})

Select the correct option(s)

(A) Molar mass of (x) is 60 $gmol^{-1}$

(B) Density of (x) is 1.25 g/cm^3

(C) Total number of unit cells of 'x' in solution are 6×10^{21} (assume entire solid is in the form of repeating unit cells i.e. undissociated)

(D) If the defected solid is found to contain 12% cationic vacancies, then % of A^{4+} ions in the crystal out of the total cations present is 40.90%.

29. Which of the following pair(s) of cations cannot be separated by using NaOH solution?

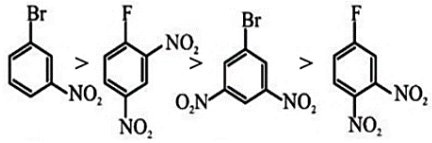
(A) Fe^{3+}, Al^{3+}

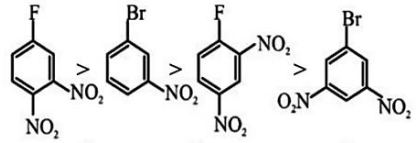
(B) Cr^{3+}, Al^{3+}

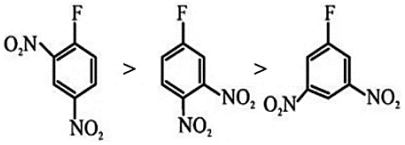
(C) Sn^{2+}, Pb^{2+}

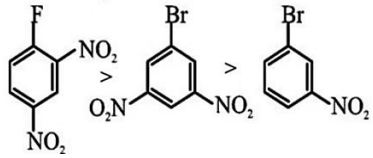
(D) Cu^{2+}, Pb^{2+}

30. Which one of the following order(s) correctly ranks the aryl halides in decreasing order of reactivity towards sodium methoxide in methanol?

(A) 

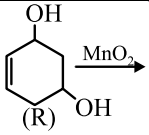
(B) 

(C) 

(D) 

SECTION-3

Numerical Value Type Questions:

31. 

In this reaction the percentage yield of the product (P) is 100%. The weight in grams of the product (P) formed by completely reacting 22.8 grams of the reactant (R) is_____.

32. For a saturated solution of AgCl at 25°C, specific conductance is 3.32×10^{-6} $ohm^{-1}\,cm^{-1}$ and that of water used for preparing the solution was 1.5×10^{-6} $ohm^{-1}\ cm^{-1}$. If the solubility product of AgCl is $x \times 10^{-10}$, then the value of x is_____.

[Given λ^{∞}_{eqv} (AgCl) = 138.3 $ohm^{-1}\ cm^2\ eqv^{-1}$]

33. The reaction $2A + B \rightarrow C + D$ goes to completion and follows the rate law $-\frac{d[A]}{dt} = k[A]^2[B]$. The sum of values of x and y in the following data will be____.

Set	$[A_0]\times10^{-3}$M	$[B_0]\times10^{-3}$M	Half-life(s)
1.	300	40	62.5
2.	300	60	x
3.	5	300	625
4.	10	300	y

34. If the total number of peptide linkages in a decapeptide is 'x' and total number of α-glycosidic linkages in a molecule of lactose is 'y', then the value of $\frac{(x)}{(y+2)}$ is _____.

35. In neutral or faintly alkaline solution, 8 moles permanganate anion quantitatively oxidise thiosulphate anions to produce X moles of a sulphur containing product. The magnitude of X is_____.

36. Consider the following reaction at 298 K

$Br_2(\ell) + Cl_2(g) \rightarrow 2BrCl(g)$

For the reaction ΔH° =25.7 kJ and the standard entropies of Br_2 (ℓ), Cl_2 (g) and BrCl(g) at the 298 K are 151.3, 221.0, 238.9 J mol^{-1} K^{-1} respectively. If ΔG° of the reaction is –x Joule then value of $\frac{|x|}{100}$ is ____.

PART-III (MATHEMATICS)

SECTION-1

Single Correct Type Questions:

37. $\bar{a}, \bar{b}, \bar{c}$ are unit vectors such that $\bar{b}$ and $\bar{c}$ are perpendicular to each other and projections of $\bar{b}$, $\bar{c}$ along $\bar{a}$ are equal, then the value of $\left|\bar{c} - \bar{b} + 2\bar{a}\right|$ is equal to

(A) $\sqrt{3}$ (B) $\sqrt{6}$

(C) $2\sqrt{2}$ (D) 2

38. Solution of differential equation

$\frac{2x^3y}{\left(1-y^2\right)}\frac{dy}{dx} = 1 - y^2\left(1+x^2\right)$ is

(A) $\frac{x^4y^2}{\left(1-y^2\right)} = 1 - cx$

(B) $\frac{x^2y^2}{\left(1-y^2\right)} = 2 - cx^2$

(C) $\frac{x^2y^2}{\left(1-y^2\right)} = cx - 1$

(D) $\frac{x^2y^4}{\left(1-y^2\right)} = cx - 2$

39. Let $f(x)$ be a positive differentiable function on $[0,a]$ such that $f(0) = 1$ *and* $f(a) = 3^{1/4}$. If $f'(x) \geq \left(f(x)\right)^3 + \left(f(x)\right)^{-1}$, then maximum value of *a* is

(A) $\frac{\pi}{12}$ (B) $\frac{\pi}{24}$

(C) $\frac{\pi}{36}$ (D) $\frac{\pi}{48}$

40. If $81(1 + 11 + 111 + \ldots\ldots$ upto *n* terms$) = [a10^{n+1} + bn + c]$ is true $\forall\ n \in N$, then

S1: $a + b + c = -18$

S2: $a^2 + b^2 = 82$

S3: $a - b + c = 10$

(A) S1, S2 are correct

(B) S2, S3 are correct

(C) S1, S3 are correct

(D) S1, S2, S3 are correct

41. Number of 4 digit numbers of the form $N = x_1\,x_2\,x_3\,x_4$ such that $2000 \leq N \leq 5000$, $4 \leq x_2 < x_3 \leq 8$ and *N* is multiple of 5 is

(A) 80

(B) 60

(C) 40

(D) 30

42. A five digit number (having all different digits) is formed using the digits 1, 2, 3, 4, 5, 6, 7, 8 and 9. The probability that the formed number either begins or ends with an odd digit, is equal to

(A) $\frac{5}{6}$ (B) $\frac{1}{6}$

(C) $\frac{1}{3}$ (D) $\frac{2}{3}$

SECTION-2

One or More Than One Correct Type Questions:

43. The solution of equation

$$\log_{\sec x}(\tan x) - \log_{\sec x}\left(\frac{\sqrt{3}}{2}\sin x\right) = 2, \text{ is } x = \lambda^\circ$$

(where $\lambda \in (0, 90)$), then

(A) Sum of square of prime factors of λ is 38

(B) Sum of square of prime factors of λ is 40

(C) Minimum value of $f(x) = \sum_{i=1}^{60}|x-i|$ is $f(\lambda)$

(D) Minimum value of $f(x) = \sum_{i=1}^{60}|x-i|$ is $f(\lambda+1)$

44. If $\mathbf{I_1} = \int_0^{\infty} \frac{x}{1+x^4}dx$,

$$\mathbf{I_2} = \int_0^{\frac{\pi}{4}} \frac{x}{\cos x(\cos x + \sin x)}dx$$

$$\mathbf{I_3} = \int_1^{\frac{1+\sqrt{5}}{2}} \frac{x^2+1}{x^4-x^2+1}\ln\left(1+x-\frac{1}{x}\right)dx$$

$$\mathbf{I_4} = \int_0^1 \frac{\sin^{-1}x}{x}dx,$$

Then which of the following is/are correct

(A) $I_1 = I_2 \neq I_4$ (B) $I_2 = I_3 \neq I_4$

(C) $I_1 \neq I_2 \neq I_3$ (D) $I_1 \neq I_3 \neq I_4$

45. Equation $x^4 + lx^3 + mx^2 + nx = -25$ has roots $\alpha, \beta, \gamma, \delta \in R^+$ then value of $l \cdot n$ can be

(A) 375 (B) 400

(C) 445 (D) 450

46. Let E_1 and E_2 be two ellipses whose centers are at the origin. The major axes of E_1 and E_2 lie along the x-axis and the y-axis, respectively. Let S be the circle $x^2 + (y-1)^2 = 2$. The straight line $x + y = 3$ touches the curves S, E_1 and E_2 at P, Q and R respectively. Suppose that $PQ = PR = \frac{2\sqrt{2}}{3}$. If e_1 and e_2 are the eccentricities of E_1 and E_2, respectively, then the correct expression(s) is/are

(A) $e_1^{-2} + e_2^{-2} = \frac{43}{40}$ (B) $\frac{e_1}{e_2} = \frac{\sqrt{2}}{\sqrt{35}}$

(C) $\left|e_1^{-2} - e_2^{-2}\right| = \frac{27}{7}$ (D) $e_1^{-2}e_2^{-2} = \frac{40}{7}$



47. Consider the equation $|x+1| - 2|x-1| = k$ then

(A) Maximum value of $|x+1| - 2|x-1|$ is 2

(B) Equation has two solutions for $k = 1$

(C) Equation has one solution for $k = 2$

(D) Equation has three solutions for $k = 3$

48. Let x_1 and y_1 be the roots of $x^2 + 8x - 2009 = 0$, x_2 and y_2 be the roots of $3x^2 + 24x - 2010 = 0$ and x_3 and y_3 be the roots of $9x^2 + 72x - 2011 = 0$. The points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$

(A) Cannot lie on a circle

(B) Form a triangle of area 2 square units

(C) Form a right angled triangle

(D) Are collinear

SECTION-3

Numerical Value Type Questions:

49. Let O be the centre of a circle of radius 1. P and Q are the points on the circle such that $\phi = \angle POQ$ is an acute angle and R is a point outside the circle such that $OPRQ$ is a parallelogram. If the area of the part of the parallelogram that is outside the circle is $f(\phi)$, then $\lim_{\phi\to 0}\frac{4f(\phi)}{\sin\phi}$ is equal to

50. Let x be the 7th term from the beginning and y be the 7th term from the end in the expansion of $\left(3^{1/3} + \frac{1}{4^{1/3}}\right)^n$. If $\frac{x}{y} = \frac{1}{12}$ then the value of n is

51. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $B = \begin{bmatrix} p \\ q \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ such that $AB = B$ and trace(A) = 50. Then det ($2A$) is

52. If complex numbers w and z satisfy the equations $|z|^2 = zw$ and $|w+\bar{w}| + |z-\bar{z}| = 8$. If w varies, then the area enclosed by locus of z is

53. Let $I_1 = \int_0^{\pi} f(x)\sin x\,dx$ and $I_2 = \int_0^{\pi} f''(x)\sin x\,dx$ (where $f(x)$ is continuous function in $[0, \pi]$). If $I_1 + I_2 = 5$ and $f(\pi) = 2$, then 10 $f(0)$ is equal to

54. If range of $f(x) = \sin\left(\tan^{-1}\left(\cos\left(\cot^{-1}x\right)\right)\right)$ is (α, β) then $\alpha + \beta$ is

PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if